

GENERAL PROJECT DOCUMENTATION

ExamTrace – Evidence-Based Examination Monitoring & Integrity Analysis System

1. Abstract

Online examinations provide flexibility and accessibility, but maintaining examination integrity in a remote environment is a major challenge. Traditional online assessment systems mainly focus on question delivery, answer submission and result calculation, while providing limited monitoring of candidate activities.

ExamTrace is a web-based examination monitoring and integrity analysis system developed to combine online assessment with automated monitoring. The system provides candidate authentication, online assessment, camera validation, webcam-based face monitoring, browser activity monitoring, violation logging, evidence capture, integrity scoring, risk classification, analytics and integrity reporting.

During an examination, the system monitors relevant events such as absence of the candidate from the camera view, detection of multiple faces, tab switching, focus loss and fullscreen exit. These events are recorded against the examination session and can be used as supporting evidence for later review.

The collected information is processed to generate an integrity score and classify the examination session according to configured risk levels. The system also provides analytical information and structured reports to assist authorized users in reviewing examination sessions.

The project is developed using Python, Flask, OpenCV, SQLite, HTML, CSS and JavaScript, with data-processing and analytical components integrated where required. The development follows an Agile approach supported by sprint planning, defect tracking and systematic testing.

2. Introduction

The growth of online education has resulted in increased use of digital examinations. Students can take assessments remotely without being physically present in an examination centre. Although online examinations provide convenience, they introduce challenges in monitoring candidate activities and maintaining examination integrity.

In a traditional examination environment, an invigilator can directly observe candidates and identify unusual activities. In an online environment, this type of physical supervision is not available. Therefore, software-based monitoring mechanisms are required to support the examination process.

A basic online examination system generally provides candidate login, question display, timer, answer selection and submission. However, these functions do not provide sufficient information about the candidate's activities during the examination.

ExamTrace addresses this requirement by combining an online examination platform with monitoring and integrity analysis capabilities. The system monitors selected webcam and browser events during an examination and maintains a structured record of the detected activities.

The recorded information is then used for integrity analysis, risk classification and report generation. This provides authorized reviewers with a consolidated view of an examination session.

2.1 Need for the System

Remote examinations require mechanisms for monitoring activities that cannot be directly observed by an invigilator.

Important events may include:

- Candidate leaving the camera view.
- Presence of more than one face.
- Switching to another browser tab.
- Losing focus from the examination window.
- Exiting fullscreen mode.
- Multiple monitoring events occurring within one session.

Recording these events systematically provides useful information for reviewing examination sessions.

ExamTrace therefore focuses on collecting examination evidence and converting relevant monitoring information into structured integrity analysis.

3. Problem Statement

Online examinations face several challenges related to monitoring and examination integrity.

The major problems are:

- Continuous physical supervision is not possible.
- Candidate presence cannot always be manually verified.
- Multiple-person presence may be difficult to identify remotely.
- Browser tab switching may not be visible to an invigilator.
- Focus-loss and fullscreen events may go unrecorded.
- Examination-related events may not be maintained systematically.
- Manual review of complete examination sessions can be time-consuming.
- Different monitoring events need to be considered together.

- Reviewers require supporting evidence rather than only a final examination score.
- Basic examination systems may not provide consolidated integrity reports.

Therefore, there is a need for an integrated examination system that can conduct assessments, monitor relevant activities, record events, collect evidence and provide integrity analysis.

4. Objectives

The main objectives of ExamTrace are:

- To develop a web-based online examination platform.
- To provide candidate authentication.
- To validate camera availability before examination.
- To monitor candidate presence using webcam-based face detection.
- To detect multiple faces during an examination.
- To monitor relevant browser activities.
- To detect focus loss and fullscreen exits.
- To record examination monitoring events.
- To capture evidence for relevant events.
- To calculate an examination integrity score.
- To classify examination sessions according to defined risk levels.
- To provide examination-session analytics.
- To generate structured integrity reports.
- To support authorized review of examination sessions.

5. Existing System

Traditional online examination platforms mainly concentrate on assessment functionality. A typical system allows a candidate to log in, view questions, select answers, complete the examination within a specified time and submit the responses.

Although this approach is useful for conducting assessments, it may provide limited information about candidate activities during the examination.

Limitations of Existing Systems

- Limited real-time examination monitoring.
- Lack of comprehensive event logging.
- Limited evidence collection.
- Difficulty identifying unusual examination activities.
- Dependence on manual supervision.
- Limited session-level analytics.
- Lack of consolidated integrity information.
- Difficulty correlating different monitoring events.

6. Proposed System

ExamTrace provides an integrated online examination and monitoring environment.

The proposed system combines candidate authentication, online assessment, webcam monitoring, browser activity monitoring, event logging, evidence capture and integrity analysis.

The overall process is:

Authentication → Camera Validation → Examination → Continuous Monitoring → Event Detection → Evidence Logging → Integrity Analysis → Risk Classification → Reporting

Major Features

- Candidate registration and login
- Candidate photo capture
- Online MCQ assessment
- Camera validation
- Face monitoring
- No-face detection
- Multiple-face detection
- Tab-switch monitoring
- Focus-loss monitoring
- Fullscreen monitoring
- Violation/event logging
- Evidence snapshot capture
- Integrity score calculation
- Risk classification
- Session analytics
- Integrity report generation

7. Scope of the Project

The scope of ExamTrace covers the complete examination monitoring process from candidate access to post-examination analysis.

The candidate can register, authenticate, complete the required verification, access the examination, answer questions and submit the assessment.

During the examination, the system monitors configured webcam and browser activities. Detected events are recorded against the examination session.

After the examination, the collected information can be processed to generate an integrity score, risk classification, analytics and an integrity report.

The system is intended to provide supporting information for authorized examination review.

8. Functional Requirements

8.1 Candidate Registration

The system shall allow candidates to register using the required information.

8.2 Candidate Login

The system shall authenticate registered candidates before providing examination access.

8.3 Candidate Photo Capture

The system shall support candidate image capture during the examination preparation process.

8.4 Camera Validation

The system shall verify camera availability before monitoring begins.

8.5 Examination Management

The system shall provide examination questions and allow candidates to select and save answers.

8.6 Face Monitoring

The system shall process webcam input to identify configured face-related conditions.

8.7 Browser Monitoring

The system shall record configured browser-related events such as tab switching and focus loss.

8.8 Event Logging

The system shall maintain records of detected monitoring events.

8.9 Evidence Capture

The system shall capture supporting evidence for applicable monitoring events.

8.10 Integrity Analysis

The system shall process recorded events to calculate an integrity score and determine the configured risk category.

8.11 Reporting

The system shall provide structured examination-session information for authorized review.

9. Non-Functional Requirements

Performance

The system should process monitoring events without significantly affecting the examination experience.

Usability

The examination interface should be simple and understandable for candidates.

Reliability

The system should maintain examination and monitoring information consistently.

Security

Candidate and examination information should be protected through controlled application access.

Maintainability

The application should be organized into manageable modules for future maintenance.

Scalability

The system architecture should allow additional monitoring and analysis functionality to be added in future versions.

10. Technologies Used

Python: Used for application logic and backend development.

Flask: Used as the web application framework.

OpenCV: Used for webcam processing and face monitoring.

SQLite: Used for storing application and examination-related data.

HTML: Used to create the structure of web pages.

CSS: Used for interface design and styling.

JavaScript: Used for client-side interactions and browser activity monitoring.

Pandas: Used for data processing and analysis where required.

Scikit-learn: Used for machine-learning-based analysis where applicable.

Matplotlib and Seaborn: Used for data visualization where applicable.

11. System Architecture

ExamTrace can be logically divided into the following layers.

Presentation Layer

This layer provides the interfaces used by candidates and authorized users.

It includes:

- Registration
- Login
- Examination interface
- Monitoring information
- Review and reporting interfaces

Application Layer

The Flask backend manages:

- Authentication
- Examination workflow
- Session management
- Monitoring events
- Integrity calculations
- Reporting

Monitoring Layer

This layer handles:

- Webcam processing
- Face detection
- Browser activity monitoring
- Focus monitoring
- Fullscreen monitoring

Data Layer

SQLite is used to store application, candidate, examination and monitoring information.

Analysis Layer

The analysis layer processes examination-session information to provide:

- Integrity scores
- Risk classification
- Analytics
- Reports

12. System Workflow

The ExamTrace workflow consists of the following stages.

Candidate Registration

The candidate provides the required information to create an account.

Authentication

The candidate logs into the system using the registered credentials.

Candidate Verification

Required candidate information and examination preparation steps are completed.

Camera Validation

The system checks whether the webcam is available and ready for monitoring.

Examination

The candidate starts the assessment and answers the questions displayed by the system.

Continuous Monitoring

During the examination, the system monitors configured webcam and browser events.

Event Detection

Relevant events such as no-face, multiple-face, tab switching, focus loss and fullscreen exit are detected.

Event and Evidence Logging

Detected events are recorded against the examination session. Supporting evidence can also be captured for applicable events.

Integrity Analysis

The collected monitoring information is processed according to the configured integrity rules.

Risk Classification

The examination session is classified using the defined scoring or classification criteria.

Analytics and Reporting

The examination-session information is processed to generate analytical information and structured integrity reports.

13. Detailed Module Description

13.1 Authentication Module

The authentication module manages access to the application.

It provides registration and login functionality and maintains the required session information.

Authentication ensures that examination functionality is available only after successful access validation.

13.2 Candidate Management Module

The candidate management module stores information associated with the examination candidate.

It manages:

- Candidate details
- Candidate authentication information
- Candidate photo information
- Examination-session association

13.3 Assessment Module

The assessment module provides the examination interface.

It supports:

- MCQ questions
- Answer selection
- Question navigation
- Timer functionality
- Answer handling
- Examination submission

The assessment module operates together with the monitoring components.

13.4 Camera Validation Module

Before the examination starts, the system validates the availability of the candidate's webcam.

The module helps ensure that the monitoring environment is ready before the assessment begins.

13.5 Face Monitoring Module

The face monitoring module uses OpenCV-based webcam processing.

It monitors the camera feed and identifies configured face conditions.

No-Face Detection

If the candidate is not detected within the camera frame, the corresponding event can be recorded.

Multiple-Face Detection

If multiple faces are detected within the monitored frame, the corresponding event can be recorded for later review.

13.6 Browser Activity Monitoring Module

The browser monitoring component records selected browser-related events.

Tab Switch Monitoring

The system records when the candidate moves away from the examination tab.

Focus Loss Monitoring

The system detects when the examination window loses focus.

Fullscreen Monitoring

The system records when the candidate exits fullscreen mode during the assessment.

13.7 Violation Management Module

Detected monitoring events are maintained as structured records.

Each record can contain information such as:

- Event type
- Session

- Time of occurrence
- Related evidence
- Monitoring information

This creates a traceable examination-event history.

13.8 Evidence Capture Module

The evidence capture component stores supporting snapshots for applicable monitoring events.

Evidence can provide additional context when an authorized reviewer examines an examination session.

13.9 Integrity Scoring Module

The integrity scoring module processes detected monitoring events and generates a consolidated score.

A conceptual representation is:

$$\text{Integrity Score} = \text{Base Score} - \text{Total Weighted Event Impact}$$

The exact event weights and scoring rules are determined by the implemented system configuration.

13.10 Risk Classification Module

The system maps the integrity information to configured risk categories.

This allows examination sessions to be grouped according to the defined risk criteria.

The classification is intended to support review and should be considered together with the recorded events and evidence.

13.11 Analytics Module

The analytics module processes examination-session information.

It can provide information related to:

- Examination sessions
- Detected events
- Integrity scores
- Risk classifications
- Session behaviour
- Event distributions

Data-processing tools can be used to organize and analyse the collected information.

13.12 Reporting Module

The reporting module consolidates examination information into a structured report.

The report may contain:

- Candidate information
- Examination information
- Session details
- Integrity score
- Risk classification
- Monitoring events
- Evidence information
- Analytical observations

14. Integrity Analysis

ExamTrace uses multiple monitoring signals to analyse an examination session.

Important monitoring events include:

- No-face detection
- Multiple-face detection
- Tab switching
- Focus loss
- Fullscreen exit

The events are associated with the relevant examination session.

The integrity analysis processes these events using configured rules and weights.

The resulting integrity score provides a consolidated representation of the recorded monitoring activity.

Risk classification is then performed using configured criteria.

The score and classification should be considered together with the underlying events and evidence when reviewing an examination session.

15. Data Analysis

The examination monitoring system produces session-level data that can be analysed to identify patterns and provide useful summaries.

Data processing can include:

- Event frequency analysis
- Session-level analysis
- Integrity-score analysis
- Risk-category distribution
- Monitoring-event analysis

Pandas can be used to process structured examination-session data.

Visualization libraries such as **Matplotlib** and **Seaborn** can be used to represent relevant analytical results.

Where implemented, machine-learning techniques such as **K-Means clustering** can be used to group examination sessions according to selected behavioural characteristics.

16. Database Design

ExamTrace uses SQLite for persistent data storage.

The database maintains information associated with different application components.

Candidate Information

Stores candidate-related information required by the system.

Examination Information

Stores examination questions, responses and submission-related information.

Session Information

Maintains examination-session details.

Monitoring Information

Stores detected monitoring events and associated session information.

Integrity Information

Stores information related to integrity scores and risk classification.

Report Information

Maintains information required for examination reporting.

The database allows information from different modules to be associated with the relevant examination session.

17. Testing and Validation

Testing was performed to verify the major functionality of the ExamTrace system.

A **Unit Test Plan containing 16 test cases** was prepared.

The test cases cover:

1. Candidate Registration
2. Candidate Login
3. Pre-assessment Camera Validation
4. No-Face Detection
5. Multiple-Face Detection
6. MCQ Answer Selection
7. Question Navigation
8. Mark for Review
9. Tab Switch Monitoring
10. Focus Loss Monitoring
11. Fullscreen Exit Monitoring
12. Integrity Score Calculation
13. Risk Classification
14. Evidence Snapshot Capture
15. Analytics Generation
16. Integrity Report Generation

Each test case contains the test procedure, condition to be tested, expected result and actual result.

The testing process helps verify that the major examination, monitoring and reporting functions operate according to their intended behaviour.

18. Defect Management

Defects identified during development and testing were recorded using a dedicated Defect Tracker.

The tracker maintains:

- Serial number
- Submitted by
- Submitted date
- Defect description
- Detected sprint
- Assigned person

- Defect type
- Action taken
- Action date
- Status
- Remarks

The defect tracking process provides traceability from issue identification to corrective action.

It also supports systematic monitoring of the development and testing process.

19. Results and Outputs

The completed system provides an integrated examination workflow.

Candidate Functions

- Registration
- Login
- Candidate verification
- Camera validation
- Examination attempt
- Question navigation
- Answer submission

Monitoring Functions

- Webcam monitoring
- Face monitoring
- Multiple-face detection
- Browser activity monitoring
- Focus-loss detection
- Fullscreen monitoring
- Event logging
- Evidence capture

Analysis Functions

- Integrity-score calculation
- Risk classification
- Examination-session analytics
- Behavioural analysis where implemented

Reporting Functions

- Session information
- Monitoring-event summary

- Integrity information
- Risk information
- Evidence information
- Structured integrity reports

20. Advantages

- Provides an integrated online examination and monitoring environment.
- Supports webcam-based candidate monitoring.
- Detects important face-related conditions.
- Records relevant browser activities.
- Maintains examination-event history.
- Supports evidence capture.
- Provides integrity scoring.
- Provides risk classification.
- Supports examination-session analytics.
- Generates structured integrity reports.
- Provides traceability for examination review.
- Reduces dependence on continuous manual observation.

21. Limitations

The performance of webcam-based monitoring can depend on camera quality, lighting and the candidate's environment.

The system requires appropriate browser and camera permissions for monitoring functionality.

Browser monitoring is limited to the events that can be detected through the implemented browser mechanisms.

Network or system interruptions may affect the online examination experience.

Automated monitoring events may not always provide complete context. Therefore, detected events and risk classifications should be reviewed with the available evidence.

The integrity score depends on the rules and weights configured within the system.

22. Future Scope

The system can be enhanced with advanced identity verification and improved computer-vision techniques.

Future versions can include more robust face-recognition models and improved detection under different environmental conditions.

Advanced behavioural anomaly detection can be introduced to identify unusual patterns across examination sessions.

The analytics dashboard can be extended with real-time charts, event timelines, session comparisons and detailed monitoring statistics.

The system can also be deployed on scalable cloud infrastructure to support larger examination cohorts.

A real-time invigilator dashboard can be introduced to display active examination sessions and important monitoring events.

Evidence management can be improved using searchable timelines and session-based evidence views.

AI-assisted reporting can also be enhanced to summarize examination-session information while keeping the underlying events and evidence available for authorized human review.

23. Conclusion

ExamTrace is a web-based examination monitoring and integrity analysis system that combines online assessment with automated monitoring and evidence-based session analysis.

The system provides candidate authentication, online assessment, camera validation, webcam monitoring, browser activity monitoring, event logging, evidence capture, integrity scoring, risk classification, analytics and reporting.

By recording multiple examination-related events within a single session, ExamTrace provides structured information that can support authorized examination review.

The project demonstrates the integration of web development, computer vision, database management, browser monitoring and data analysis in an online examination environment.

The development process was supported by Agile planning, defect tracking and systematic functional testing, including a 16-case Unit Test Plan.

24. References

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